Course announcement: 246C, complex analysis

Terence Tao · 13 March 2018

https://terrytao.wordpress.com/2018/03/13/course-announcement-246c-complex-analysis/

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Next quarter (starting Monday, April 2) I will be teaching Math 246C (complex analysis) here at UCLA. This is the third in a three-series graduate course on complex analysis; a few years ago I taught the first course in this series (246A), so this course can be thought of in some sense as a sequel to that one (and would certainly assume knowledge of the material in that course as a prerequisite), although it also assumes knowledge of material from the second course 246B (which covers such topics as Weierstrass factorization and the theory of harmonic functions).

246C is primarily a topics course, and tends to be a somewhat miscellaneous collection of complex analysis subjects that were not covered in the previous two installments of the series. The initial topics I have in mind to cover are

- Elliptic functions;
- The Riemann-Roch theorem;
- Circle packings;
- The Bieberbach conjecture (proven by de Branges); and
- the Schramm-Loewner equation (SLE).
- This list is however subject to change (it is the first time I will have taught on any of these topics, and I am not yet certain on the most logical way to arrange them; also I am not completely certain that I will be able to cover all the above topics in ten weeks). I welcome reference recommendations and other suggestions from readers who have taught on one or more of these topics.

As usual, I will be posting lecture notes on this blog as the course progresses.

[Update: Mar 13: removed elliptic functions, as I have just learned that this was already covered in the prior 246B course.]

Comments (8)

Anonymous · 13 March 2018

Another possible topic is H^p spaces.

Maths student · 15 March 2018

I can't wait for the notes, and will certainly gladly give any corrections I might find.

Jaikrishnan Janardhanan · 16 March 2018

One topic I am very fond of teaching is the Ahlfors–Schwarz lemma and using to give proofs of the big Picard theorem, the general Montel's theorem, etc. The book "Complex Analysis: The geometric viewpoint" has a very nice presentation of all this. The book also introduces several invariant metrics which are indispensable in the study of holomorphic mappings in higher dimensions.

John Mangual · 18 March 2018

There's a discussion of Riemann Roch theorem in Chapter 21 of Fulton's "Algebraic Topology" after the chapter on Riemann Surfaces and Algebraic Curves in Part X on Riemann Surfaces. It is also in Chapter 5 of Joe Polchinski's book on String Theory.
One wishes to integrate objects over a "curve" $X = \{(z, w) : F(z, w) = 0\}$. I very modestly assume that "cohomology" just means "integrals" and carry on.
I don't know much about SLE except that it relates conformal mappings and random walk. It gets very complicated. Oded Schramm wrote about many exciting statistical models during his time at Microsoft :-)

Rob · 20 March 2018

Are the notes from 246B available? Thanks :)

> **Russell** · 20 March 2018
>
> John Garnett taught the 246B course, we did the last few chapters of Ahlfors Complex Analysis, Caratheodory's theorem, the Uniformization theorem, and intro to univalent functions.
>
> > **Rob** · 20 March 2018
> >
> > Thank you Russell :)

Anonymous · 25 March 2018

It would be interesting to see some elliptic modular forms in the course.